# An Evaluation System for Building Footprint Extraction From Remotely Sensed Data

Chuiqing Zeng, Jinfei Wang, *Member, IEEE*, and Brad Lehrbass

***Abstract*—This study proposes a multi-criteria and hierarchical evaluation system for building extraction from remotely sensed data. Most of current evaluation methods are focused on classification accuracy, while the other dimensions of extraction accuracy are usually ignored. The proposed evaluation system consists of three components: 1) the *matched rate*, including evaluation metrics for the traditional classification accuracy (e.g., *completeness, correctness, and quality*); 2) the *shape similarity* that describes the resemblance between reference and extracted buildings, including image-based and polygon-based metrics; and 3) the *positional accuracy* which is measured by distances at feature points such as building's centroid. The system also hierarchically evaluates extracted buildings at *per-building*, *per-scene*, and *overall* levels. To reduce the redundancy among different metrics, principal component analysis and correlation analysis are employed for metrics selection and aggregation. Four different building extraction methods, using high-resolution optical imagery and/or LiDAR data, are implemented to test the proposed system. The experiment demonstrates that the proposed system is more consistent with human vision compared to traditional classification accuracy metrics. This system can highlight perceptible differences between the extracted building footprints and the reference data, even if this difference is insignificant measured by the traditional metrics.**

***Index Terms*—Building footprint accuracy evaluation, high spatial resolution imagery, LiDAR, object-based building extraction, positional accuracy, shape similarity, relief displacement correction.**

## I. Introduction

**B**UILDING footprint detection from remotely sensed data is of great importance to disaster (earthquake, flood or fire) management, real estate industry, homeland security and many other applications [1]. Building footprint extraction is a well-developed topic that has been explored in many studies [1]–[3]. Current automatic method for building extraction, however, is impossible to reach a perfect 100% accuracy. The main reasons behind the deficiency are scene complexity, incomplete cue extraction and sensor dependency [4]. Considering the large number of extraction methods but lack of standard techniques to evaluate them, the evaluation methods become very important. Although there are several studies [1], [5]–[9] about the evaluation of building extraction, most of them still concentrate on the image classification accuracy, rather than a framework to thoroughly evaluate the extraction results. In one study an evaluation system is defined [10]; however, there is no experiment to empirically verify the system. Another problem with the evaluation criteria is the redundancy of metrics. Most studies list related metrics (e.g., 15 metrics in [1]), but cannot provide a clear picture about the performance of a given extraction method.

This study provides a brief review of building extraction methods, and a detailed review of current evaluation metrics for building extraction. The objective of this study is to organize current building evaluation metrics, to reduce the redundancy, and to develop an evaluation system for building extraction accuracy assessment.

## II. A Brief Review of Building Extraction Methods

Aerial photographs and high-resolution optical satellite images are typical data sources for the extraction of building's boundaries. Various methodologies have been developed based on the conceptions and characteristics of buildings, including edge detection, classification, Hough transformation, seed growing, geometric, photometric and structural analyses, and spectral, structural and contextual analyses [11]. In order to tackle the complicated context of buildings in urban areas, other methodologies such as snake model [12], [13] and energy function [14], have been introduced to building footprint extraction. Recently, buildings have been extracted directly from 3-D environment by a template matching technique for height estimation and potential building segment detection [15]. Optical images alone, however, can only obtain limited spectral information of ground features in visible bands. It is difficult to differentiate ground features with similar spectral characteristics (e.g., buildings and other man-made features). Spatial information from the high-resolution images increases the separablility between buildings and non-buildings but still cannot resolve the spectral confusion. As a result, auxiliary information, such as vector parcel geometries and their attributes from Geographic Information Systems (GIS) [11] or high-resolution interferometric SAR (InSAR) data [16], can benefit and improve the building extraction.

In recent decades, the advent of light detection and ranging (LiDAR) data has challenged the conventional building footprint extraction. The ability to detect distances [17], as well as the high spatial resolution, gives airborne LiDAR the advantage

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to extract building footprints accurately. LiDAR can be used on its own to extract building footprints via height, size, and shape information [18]. By tracing and regularization of building boundaries from raw LiDAR point clouds, some studies separate building and non-building LiDAR points via slope [19] or morphological filtering [20], segmenting LiDAR points that belong to the same building, tracing building boundary points, and regularizing the boundaries. These studies report high accuracy on building footprint extraction. Furthermore, LiDAR is even applied to roof plane segmentation and roof model reconstruction via level set approaches [21].

Finally, optical imagery and LiDAR are integrated for more accurate building footprint extraction. LiDAR data contains height and intensity but does not have color information, while optical imagery has color bands but is lack of height information. As a result, these two complementary data are combined for better recognition of buildings [22]. In some studies, LiDAR is used to extract coarse building boundaries and then overlaid with optical image to retrieve precise building boundaries [22], [23]. In some other studies, a LiDAR point cluster is recognized as an isolated building object by its similar height property and homogenous normalized difference vegetation index (NDVI) derived from optical imagery [1], [4]. A recent effort aims to separate buildings from trees in complex scenes with hilly and dense vegetation [24].

Another trend in building footprint extraction is the improvement of spatial resolution for optical imagery being used. This trend is accompanied by the increasing impact of relief displacement on aerial images [25] and its considerable effect on building boundary extraction. Currently, little attention has been paid to correct this displacement [26]–[28]; few studies have taken true ortho-rectification into consideration before extracting building footprint. Our previous work had developed a method to correct the relief displacement of tall objects from the already ortho-retificated aerial images [29].

The current building footprint extraction methods are mixtures of different data sources, various algorithms, as well as different degrees of human interference. Most studies propose methods and evaluate their results using diverse criteria. The inconsistency of evaluation methods makes it difficult to compare various approaches; therefore, a universal evaluation system is needed.

## III. A Review of Evaluation Methods for Extracted Buildings

### A. Matched Rates

Building footprint extraction results have been evaluated by different strategies in many studies. The most popular evaluation method is transplanted from traditional image classification accuracy assessment [30]. In an evaluation, two classes are considered: buildings (or the object) and the background. The extracted objects are evaluated against the reference objects. As a result, four different parameters [31], true positive (TP, the common area of extracted objects and reference objects), false positive (FP, the area of extracted objects but not reference objects), false negative (FN, the area that belongs to the reference but not the extracted result), and true negative (TN, the area that belongs to neither the extracted result nor the reference), are derived from the reference image and the building extraction image. Different metrics are developed for evaluation as defined in Table I.

TABLE I
Different Metrics Derived From the Traditional Image Classification Evaluation

| Metrics | *Completeness* | *Correctness* | *Quality* | *Kappa** |
|---|---|---|---|---|
| Definition | $\frac{TP}{TP+FN}$ | $\frac{TP}{TP+FP}$ | $\frac{TP}{TP+FN+FP}$ | $\frac{n\times(TP+TN)-pr}{n^2-pr}$ |
| Metrics | *Omission error* | *Commission error* | *Branching factor* | *Miss factor* |
| Definition | $1-\frac{TP}{TP+FN}$ | $1-\frac{TP}{TP+FP}$ | $\frac{FP}{TP}$ | $\frac{FN}{TP}$ |

*in Kappa, n=TP+FP+TN+FN, and $pr=(TP+FP)(TP+FN)+(TN+FP)(TN+FN)$

The *Completeness*, also known as *producer's accuracy* [30] or *detection rate* [10], represents the percent of the buildings being correctly detected with respect to the reference data; while the *Correctness*, referred to as *user's accuracy* [30] and *overlap* [8], describes the percent of correctly detected part to the extracted buildings. On the other hand, *Omission error* [10] is the percent that is not detected from the reference, while *Commission error* [10] is the incorrectly detect part within the extracted buildings. *Quality* [5], [7], [31], similar to *fitness* [8], is a metric that combines the *Completeness* and *Correctness*. *Kappa coefficient* [9] takes into account the agreements contributed by chance. *Branching factor* and *Miss factor* [5], [31] describe two types of possible mistakes in the extraction: the former indicates the rate of incorrectly labeled building pixels, the latter gives the rate of missed building pixels. *underlap* and *extralap* [8] correspond to *Branching factor* and *Miss factor*, but they are slightly different on the definition. The metrics listed are originally used in the image classification at pixel level, they are later adopted in feature-based applications [1], [9]. Metrics in the second row in Table I with higher values indicate better extraction performance; metrics in the fourth row with lower values correspond to better performance. Abbreviations are used: *Comp* denotes *Completeness, Corr* denotes *Correctness*, $Omiss_e$ is *Omission error*, $Comm_e$ *is Commission error, Q* represents *Quality*, *Mf* is *Miss factor*, *Bf* is *Branching factor*, and *Kappa represents Kappa coefficient*.

### B. Shape Similarity

Shape similarity is another category of metrics which describes to what extent an extracted object is similar to the reference with respect to their shapes. The *shape similarity* is a subjective and not well-defined problem. For a given reference and an extracted building, the only certain status is the perfect match when the extracted shape is exactly the same as the reference. In current literature, two types of strategies have been developed to measure objects' similarity. One is to evaluate similarity of the boundary (or "contour") between an object and a reference; another strategy compares an object and its close neighbors at image level with image related similarity.

On the boundary level, topological relations such as overlapping, disjoint, and covering for each pair of objects in reference and extracted maps are used to define a metric of similarity [32].

TABLE II
THE METRICS FOR OBJECT-BASED *SHAPE SIMILARITY*

| metrics | *Area difference* | *Perimeter difference* | *Mean* (r(*Area*)) | *Std* (r(*Area*)) |
|---|---|---|---|---|
| definition | $\left[\dfrac{abs(Ae-Ar)}{Ar}\right]_i$ | $\left[\dfrac{abs(Pe-Pr)}{Pr}\right]_i$ | $Mean\left[\left(\dfrac{\min(Ae,Ar)}{\max(Ae,Ar)}\right)_i\right]$ | $Std\left[\left(\dfrac{\min(Ae,Ar)}{\max(Ae,Ar)}\right)_i\right]$ |

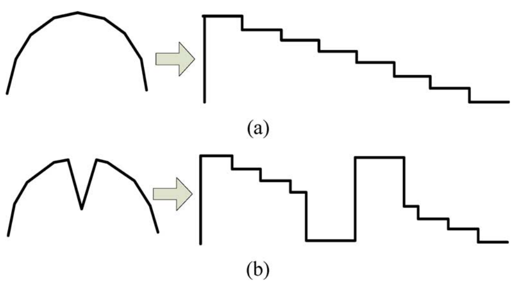

Fig. 1. The reference and extracted curves and their tangent functions. (a) The reference curve, with the curve on the left and its tangent function on the right. (b) The extracted curve and its tangent function.

*Corner positional difference*, *Perimeter difference* and *Area difference* are defined in [10]. These *differences* describe the absolute (corner, perimeter, and area) differences from extracted buildings to the reference, divided by the reference. In contrast to *difference*, the *ratios* of the above measures between the extracted building to the reference building [6] are another way to describe similarity. To ensure that the *ratio* is between 0 and 1, the ratio of the minimum to the maximum of a given shape descriptor (i.e., *ratio* of Area is referred to as *r(Area)*), is designed [9]. Different similarity measures at the building boundary level are summarized in Table II.

Another metric "Tangent Space Representation (TSR)" [33] is developed to evaluate the object boundary by comparing their outline difference step by step. For each line segment in a shape, the derivative of the segment, called "tangent function", is calculated and connected consecutively to form a line. As shown in Fig. 1, two shapes are displayed on the left: the reference in Fig. 1(a) and the extracted shape in Fig. 1(b), with the generated tangent function on the right, respectively. The lengths of the two shapes are rescaled to 1, respectively, and the tangent function difference between these two lines is used to measure the similarity.

On the image level, *Moment-Derived* Shape Similarity [10] is designed to measure the similarity. That is because geometric moments can provide an equivalent representation of an image in the sense that an image can be reconstructed from its moment. The moment-based similarity is the Euclidean distance in a space defined by the first two moments [34] as the two axes. In addition, in [35], mutual information (MI) and its derivatives, *Joint entropy*, *Entropy correlation coefficient*, and *Correlation ratio* are used as image similarity metrics, with most of them reported to have similar performances.

### C. Positional Accuracy

*Positional accuracy*, also referred to as *geometric position accuracy* [1] or *location accuracy*, is another group of metrics that evaluate the geometric position difference between corresponding points of the extracted objects and the reference. Because of the limitations on current data sources and methodology, together with perceived unimportance to applications, positional accuracy is rarely mentioned in current evaluation systems. Root mean squared error (RMSE) is used to measure the positional errors between points in the reference and extracted maps [10]. Usually the corner points of buildings are used as the points for evaluation. Furthermore, in [9], Euclidean distance between the centers of the mass of an extracted object and the corresponding object is used as another metric to measure positional accuracy. Both *mean* and *standard deviation* are useful for evaluation.

In summary, evaluation strategies for building extraction methods from current literatures are summarized as three different categories based on their evaluation purpose: *matched rate*, *shape similarity*, and *positional accuracy*. There are other ways to group those evaluation methods too. *Pixel-based* and *object-based* evaluation methods are commonly adopted in many studies [1], [7], [36]. Evaluation methods labeled as pixel level or object (or "building") level use different spatial data models: a raster image or vector polygons; however most metrics can be calculated in either data model. Moreover, the evaluation methods can also be grouped as *local* and *global* [6] from the perspective of scales. The *local* level refers to the single object level, while the *global* level analyzes all objects in the entire image. Finally, the evaluation system can be built at different levels. Three levels (*number-based, area-based*, and *shape similarity indices* [10]) of the indices are used depending on how rigorous the requirements of accuracy assessment are for the desired application.

The problem with the current evaluation metrics can be summarized as: 1) most evaluation methods concentrate on the traditional classification accuracy, described by metrics such as *Correctness*, *Completeness*, and *Quality*; the other two components of the evaluation, the shape similarity and positional accuracy, have not attracted enough attention. 2) The shape similarity component, primarily described by area and perimeter ratio is not well developed; it needs further systematic and thorough analysis to be used to describe the similarity of complex shapes. 3) Various metrics are reported in many studies; however, they are not well-organized. It is difficult for users to assess a method with more than ten metrics at the same time. 4) The correlation between different metrics has not been investigated.

Strong correlation between some of the reported metrics indicates redundant information and conceals the nature of evaluation methods.

## IV. A Multi-Criteria Evaluation System

### A. Assumptions

The evaluation process can be divided into two steps. The first step is the registration of buildings between reference data and the extracted results; the second step is to compute metrics and conduct evaluation. There are efforts involved automatic implementation and multi-detection consideration [37] in the registration step. We acknowledge the importance of such efforts because they facilitate the evaluation process; however, the focus of this study is on the second step regarding how to develop an evaluation system. To clarify the expression for connected buildings and avoid topological mismatch between reference and the extracted buildings, a semi-automatic registration method is used in this study. The building registration is automatic with one-to-one (including zero-to-one and one-to-zero) relations between the extracted buildings and reference buildings. Many-to-many relations occur infrequently and only when there are topological mismatch between reference buildings and the extracted buildings (19 out of 761 buildings in this study). Many-to-many relations are clarified by a manual method which applies the topological clarification [7] via splitting and merging extracted buildings.

The definition of TP is important because TP directly relates to the classification accuracy. There are different ways to calculate TP. One popular approach is setting a threshold for the percent of overlap between the extracted and reference buildings [9], [36]. In another study [10] no threshold is set: as soon as an overlap between the two buildings is detected, that building will be used for validation. Considering that the threshold is subjective, in [7] a Point-in-Polygon test is used to determine TP. In this study, TP definition is adopted from [10] and no percent threshold is set if overlap exists. A minimum size of a building (4 $m^2$) is set to avoid obvious mistakes.

### B. The Demand for a Multi-Criteria Evaluation

Although the popular *matched rate* is effective to evaluate the building's matching accuracy at pixel level, *matched rate* alone is not sufficient for a complete evaluation for buildings. As shown in Fig. 2, there are three different extracted buildings compared with the reference. In Fig. 2(b), the courtyard is displaced to two possible locations: A or B; case A and B lead to the same *matched rate* result, but they are two different shapes. In (c), a building is rotated because of image registration problem although it has the same shape as that in the reference; the *matched rate* is quite low for this matching. In (d), a building is extracted with similar size and location but with irregular boundaries; although this extraction will give a high *matched rate*, the perimeter is significantly larger because its jagged boundaries and the shape will be quite different if it is used in 3D reconstruction. Also, the corner positions are changed in this case.

The demand of a multi-criteria evaluation system is twofold. On one hand, the commonly used user's and producer's accuracies are not able to completely evaluate the performance of building extraction, because other aspects (e.g., the *shape similarity* and *positional accuracy*) are not effectively described. Performance on the other aspects is increasingly concerned in applications that demand more building details such as detailed 3D reconstruction. On the other hand, the accuracy of extracted buildings has increased remarkably with advanced methods and higher-resolution imagery in recent years. For example, recent extraction methods can achieve the traditional user's and producer's accuracy at 90% level [38], [39], which leaves limited room for further comparison of those methods. Therefore, a multi-criteria evaluation is expected to meet both demands.

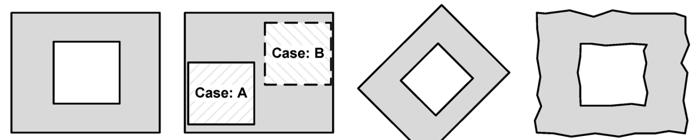


Fig. 2. Hypothetical examples of different building extraction results compared with the reference. (a) Reference building boundary. (b) Extracted result 1: an extracted building with the courtyard displaced to A or B. (c) Extracted result 2: a rotated building. (d) Extracted result 3: a polygon with irregular borders.

### C. The Selection of Metrics for Evaluation

Among all the proposed *matched rate* metrics, connections can be derived from their definitions. From Table I, the connection between different metrics built from TP, FP, and FN are summarized as follows:

- $Comp + Omiss_e = 1$; $Corr + Comm_e = 1$;
- $Q = \dfrac{Comp \cdot Corr}{Comp + Corr - Comp \cdot Corr}$; $Q = \dfrac{1}{Bf + Mf + 1}$;
- $\dfrac{1}{Omiss_e} + 1 = \dfrac{1}{Mf}$; $\dfrac{1}{Comm_e} + 1 = \dfrac{1}{Bf}$

From the listed relations, *Comp* and $Omiss_e$ describe the two sides of a coin; so do the *Corr* and $Comm_e$. $Omiss_e$ relates to *Mf*, while $\mathrm{Comm}_e$ and *Bf* share the same relation. Furthermore, $Q$ is a metric by integrating both *Comp* and *Corr* in its definition. Considering the relations between $Q$ and other metrics directly or indirectly, $Q$ represents most of the above metrics. In this study, *Mf, Bf*, $Comm_e$, and $Omiss_e$ are discarded to avoid redundant information, while *Comp, Corr* and $Q$ are computed for further analysis. *Kappa* is a metric which is more proper to calculate at image level for all buildings.

Apart from the *shape similarity* metrics mentioned in the previous review (Section III), many shape indices are designed in eCogontion to measure an object's shape [40]. These indices include *Area, Perimeter (Peri), Border index (bdr_idx), Asymmetry (Asymm), Density (Dst), Compactness (Cmpt), Length/width (L/W), Main direction (Dir), Elliptic fit (Elp_fit), Rectangle fit (Rect_fit)*, *shape index* (*Shp_idx*) and *Roundness* (*Rnd*). *Area* and *Perimeter* are the primary metrics to measure a shape. The *Asymmetry* describes the relative length of an image object, compared to a regular polygon. *Compactness* is calculated by the product of the length and the width, and divided by the number of pixels in an object. *Length/width* describes whether

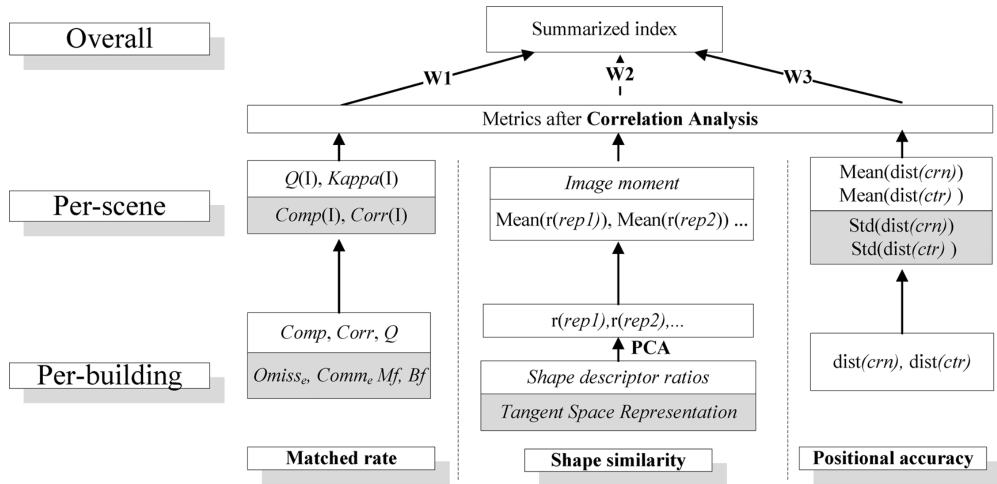


Fig. 3. A multi-criteria and hierarchical evaluation system for building extraction. The system consists of three components (see columns): *matched rate*, *shape similarity* and *positional accuracy*; it also has three levels (rows), at *per-building*, *per-scene*, and *overall* levels, respectively.

the object is close to a square (when it equals to 1). *Main direction* represents the major direction of the object. *Elliptic fit* and *Rectangular fit* calculates to what extent the object can be fitted in an ellipse or a rectangle. The *Roundness* describes how similar an object is to an ellipse.

These shape indices can be easily converted to *shape similarity* metrics by computing the *ratio* between an extracted building and a reference building, according to the definition in Table II. To evaluate the *shape similarity*, in this study metrics for all the indices are firstly computed and then representative metrics are selected. "Tangent Space Representation" is not implemented in this study because experiments show that it is sensitive to details on the boundaries and it is complicate to compute with inner rings of polygons.

Finally, for the *Positional accuracy* evaluation, the distances from corresponding check points between a reference and extracted building are measured. Such check points can be corner points or centroids: the distance at corresponding corner points is denoted as "dist(*crn*)"; the distance for corresponding centroids is referred to as "dist(*ctr*)".

### D. A Multi-Criteria and Hierarchical Evaluation System

A multi-criteria evaluation system is built hierarchically with three levels (Fig. 3): The *per-building* level at bottom describes metrics for each single building. The *per-scene* level in the middle describes each metric for a whole scene. The *overall* level defines a summarized index. Currently most evaluation methods are developed at *per-scene* level and provide metrics for the entire study area; the proposed *overall* level aims to provide a single index for effective comparison of building extraction results. The *per-building* metrics can provide detailed information about the performance of a certain extraction method on individual buildings. It is useful in large scale mapping concerning the accuracy of each building. Detailed metrics provide valuable information for further manual editing and correction of each building from different perspectives.

The evaluation system, shown in Fig. 3, is also divided into three different components as described in Section III. For the *matched rate* component, *Comp*, *Corr* and $Q$ are computed at *per-building* level. Then *Comp(I)*, *Corr(I)*, $Q(I)$ and *Kappa(I)* are calculated for all buildings in the whole image at *per-scene* level (e.g., $Q(I)$ means the *Quality* for the image). For the *shape similarity* component, ratios of shape measures from the reference and extracted buildings are used. The *image moment* [10] similarity (distance metric) is calculated at the *per-scene* level. For the *positional accuracy* component, distances on corner points and polygon centroids are used. The corner points use ground surveyed points as reference. When ground survey is unavailable, corresponding corner points from reference buildings instead can be treated as an alternative.

Two analysis methods are employed to reduce the redundancy between metrics. Firstly, within *shape similarity* metrics, a principal component analysis [41] is conducted to select representative metrics, denoted in Fig. 3 as "*Rep1*", "*Rep2*", etc., which are used in the upper levels. Secondly, for all metrics at *per-scene* level, a correlation analysis is performed to remove highly correlated metrics between different components. Moreover, this system may be extended to evaluate 3-D building extraction in the future. For example, for *matched rate*, the area on 2-D plane can be replaced by 3-D volume of the building and/or the surface area; for *shape similarity*, building's projection on the three-dimensional planes can replace the current 2-D plane for metrics' computation; for *positional accuracy*, distance can be directly used in 3-D space.

To generate a final summarized index, different weights for the three components may be set by users according to specific applications. Although weights for the three components are subjective, an integrated and summarized index is easier to use for comparison purposes. The summarized index (*Sum_Idx*) can be computed as follows:

$$Sum\_Idx = Mated\_rate \times w_1 + Shape\_similarity \times w_2 + Positional\_accuracy \times w_3 \quad (1)$$

where $w_i$ $(i = 1, 2, 3)$ is the weight.

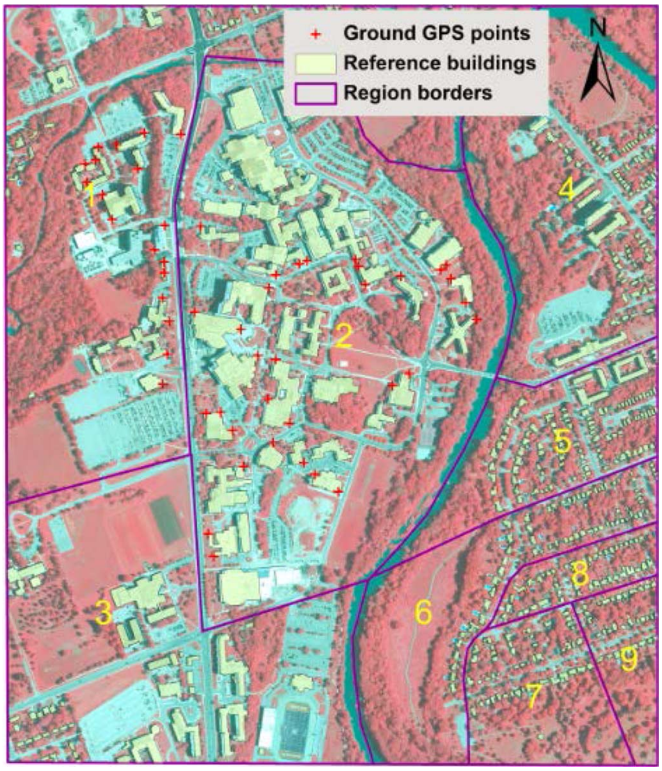


Fig. 4. The study area with reference buildings, ground GPS points and subregions. The background is the CIR imagery.

## V. Experiments for Building Extraction Methods

### A. Study Area and Data Sources

The study area is located at the campus of the University of Western Ontario (UWO) in London, Canada, and its surrounding residential areas, as shown in Fig. 4. The study area includes various land cover types: impervious surface, trees, rivers and bare grounds. Building footprint sizes also vary significantly in the study area according to building functions. There are tall buildings on the campus of UWO and low residential buildings nearby.

LiDAR data and aerial Color-Infrared (CIR) imagery were collected for the study area. The LiDAR data, with an original density of 0.8–0.9 points/m$^2$ is re-sampled and interpolated to a 1 m resolution raster image. The optical image is CIR imagery with 0.3 m spatial resolution captured in 2008 by the City of London for a tree cover mapping project [29]. The CIR images include the green, red and near-infrared bands. The reference building footprint vector file comes from the City of London (2006), while the ground reference GPS points were surveyed in October, 2011 using a differential GPS with absolute position accuracy within 1 meter. The GPS points only cover the campus area because most residential buildings are not accessible for measuring. For comparison and analysis purpose, the study area is divided into nine different regions with roughly equal building numbers in each region. The roads and rivers are used as region borders. Regions 1 to 3 are mainly public service buildings, while Regions 4 to 9 are mainly residential buildings.

### B. The Four Different Building Extraction Methods

Four different methods for building extraction in the study area have been conducted as shown in Fig. 5. The first three methods use the original CIR image, LiDAR and their combination, respectively. The fourth method, developed by our previous work [42], corrected the relief displacement in the already ortho-rectified CIR image before combining it with LiDAR for building extraction. For details about the relief displacement correction, please refer to our papers [29], [42]. The four methods are:

1) The CIR image method (denoted as "CIRimage"), which uses the spectral information alone and extracts buildings via object-based supervised classification. This is the most straightforward method to extract building footprint from optical imagery.
2) The LiDAR method (denoted as "LiDAR"), which relies on the normalized DSM (nDSM), together with the height and object's shapes for extraction. To distinguish buildings from other high objects, their shape and relationship to surrounding high objects are also utilized for extraction.
3) The method that combines LiDAR data with the original CIR image before the relief displacement correction (denoted as "LiDAR+unCorr"), which exploits both height and color information. The color information from the CIR images and the height information from the LiDAR data are combined in the object-based algorithm to develop a rule set. A better extraction result is expected.
4) The method that combines LiDAR data with the CIR image after the relief displacement correction (denoted as "LiDAR+Corr"), which aims to investigate the effect of relief displacement on building extraction. In an ortho-rectified CIR images based on a digital terrain model, tall buildings are still misaligned due to relief displacement. Based on the nDSM from LiDAR, the cross-track relief displacement can be predicted for all points on the ortho-image [29], [42]. With this method, the tall objects can be aligned between the LiDAR and the CIR images. The object-based rule set is the same as in the "LiDAR+unCorr" method.

Object-based segmentation and classification is used for Methods (1), (3), and (4) and a decision tree is developed for these methods. Method (2) is implemented by commercial software for airborne LiDAR. The results from the four methods are exported as vector maps and are further refined. To facilitate the comparison process, each building in the reference map is given a joint ID to be connected with a building in each extracted result. An overlap operation between reference and the extracted result can automatically detect simple "one-to-one" relations; for buildings with "many-to-one" relations, topological clarification [7] is carried out to detect and process topological mismatch. An extracted building covering more than one reference buildings is split, while two or more extracted buildings inside the same reference building are merged. For "one-to-zero" and "zero-to-one" relations between extracted and reference buildings, no *shape similarity* is calculated.

The *Area* and *Centroid* for each building in the extracted and reference maps are computed; together with the *overlap Area*,

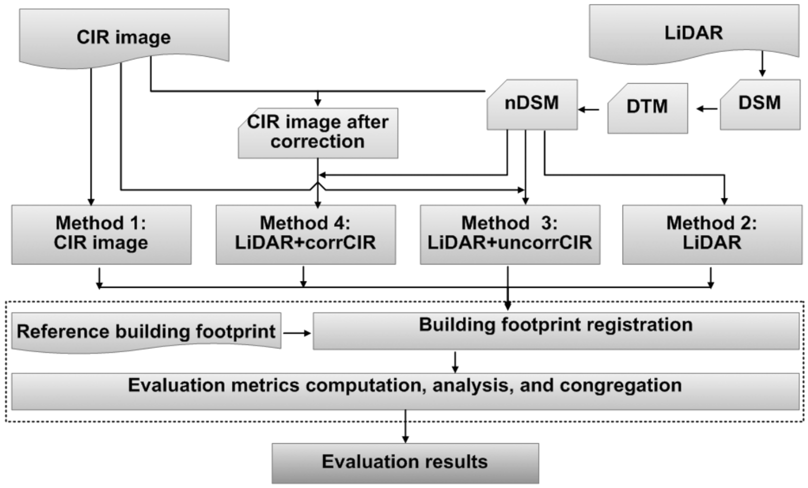


Fig. 5. The procedure of performance evaluation of the four methods for building footprint extraction.

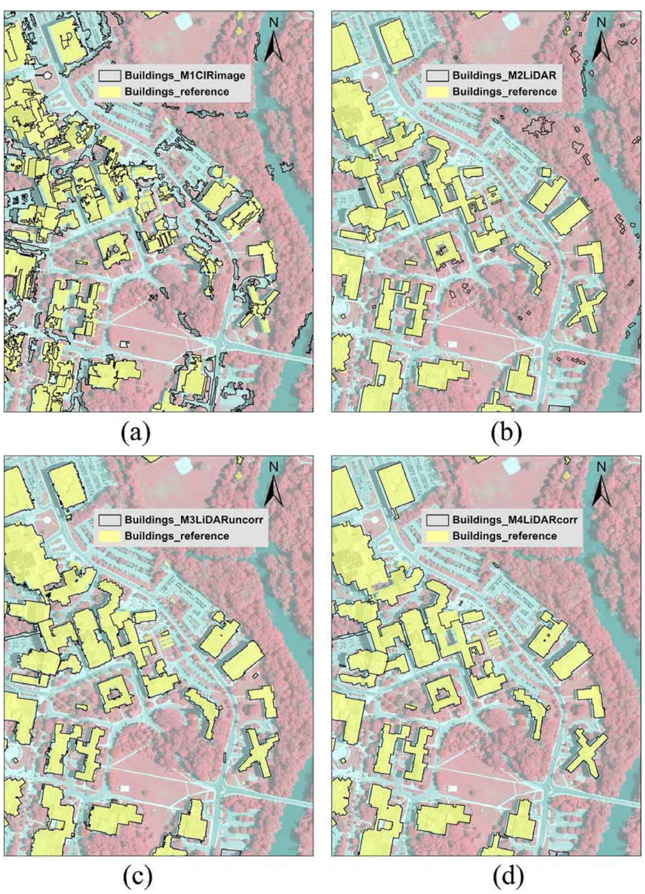


Fig. 6. A subarea showing the four building extraction results. (a) The extracted result from Method 1—CIR image (M1). (b) The results from Method 2—LiDAR (M2). (c) The result from Method 3—LiDAR and CIR image before relief correction (M3). (d) The result from Method 4—LiDAR and CIR image after relief correction (M4).

*TP*, *TN*, *FP*, and *FN* are calculated. The corner points of the extracted buildings are identified by matching the corresponding corner points to the GPS survey points. *Shape similarity* metrics on polygons are computed by commercial software and saved as attributes for each polygon. The image moment based distance is computed between the rasterized images of the extracted and reference buildings.

### C. Building Extraction Result Maps

From Fig. 6, the method M1 with CIR image presents the worst result, and it extracts all the buildings with random shapes. Another problem is that buildings are confused with parking lots. M2 with LiDAR data provided a much better result. However, M2 has some extraction errors along the rivers in the densely treed areas. This is because LiDAR does not have color information and buildings can only be extracted based on height and shape. In densely tree covered areas and with shapes similar to buildings, errors will occur. From visual evaluation, the results for M3 and M4 look similar. Both of them have high accuracy with all buildings extracted and non-building objects removed. However, when looking into details, the extracted building boundaries in M4 are "cleaner" with less unnecessary vertexes. Further analysis will be discussed based on computed metrics in Section VII.

## VI. Selection and Summary of Metrics at Different Levels

### A. Redundancy Reduction in Shape Similarity Based on Principal Component Analysis

When comparing the reference and extracted buildings at *per-building* level, twelve possible *shape similarity* ratios are used as listed in Section III. Based on the result from M4 ("LiDAR+Corr"). A Principal Component Analysis (PCA) is performed on these ratios and the first four Principal Components (PC) of the PCA are shown in Fig. 7. The biplot in Fig. 7(a) shows the coefficients of the first two principal components (PC1 and PC2) and Fig. 7(b) displays the biplot for the third and fourth components (PC3 and PC4). The magnitude and sign of each shape metric's contribution to the principal components are shown as the length and direction of 2-D vectors, while the samples are shown as dots. For example in the biplot Fig. 7(a), the ratio for *main direction* vector has

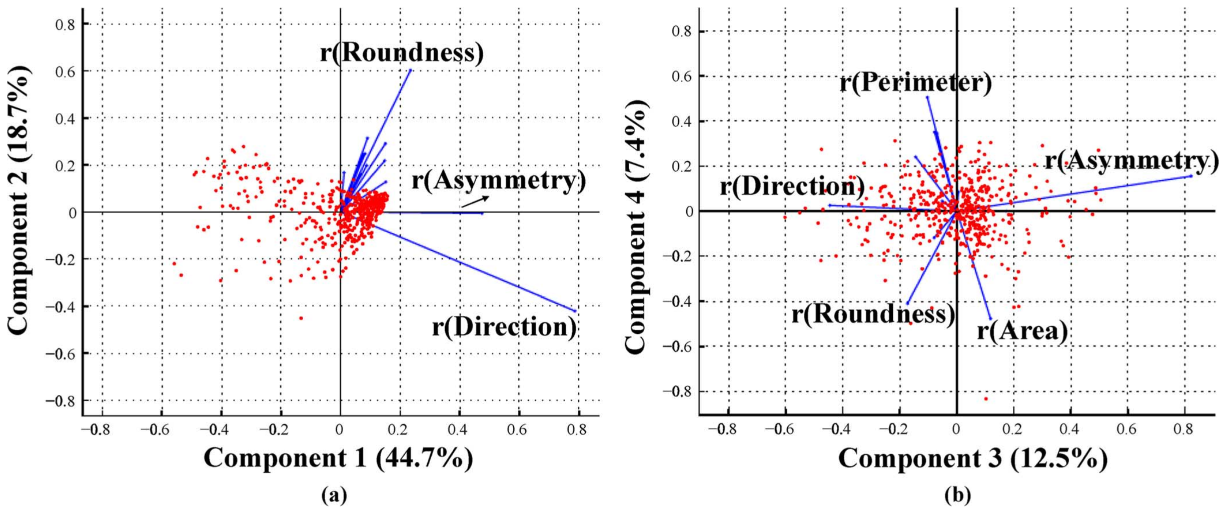


Fig. 7. The principal component analysis for twelve *shape similarity* metrics. (a) The biplot of the principal components 1 and 2 with samples. The coordinates for a ratio metric indicates its contribution to components PC 1 and PC 2 respectively. The percent in the axis label indicates the amount of variance accounted for by that component. (b) The corresponding biplot for PC 3 and PC 4.

coordinates (0.8, $-0.42$), because its correlation coefficient to PC 1 is 0.8 and that for PC 2 is $-0.42$.

From Fig. 7, PC 1 accounts for 44.7% of variance and the major contribution comes from *Direction*, *Asymmetry*, and *Roundness*; PC 2 covers 18.7% of variance and it is contributed by *Direction* and *Roundness*. PC 3, with 12.5% of variance, is mainly influenced by *Asymmetry* and *Direction*; and PC 4, with only 7.4% of variance, is related to *Area*, *Perimeter* and *Roundness*. The accumulated variance from the first four components is about 85% of the total variance. Experiments on the other three methods (M1 to M3) report similar results. Based on the above analysis, the *shape similarity* metric at *per-building* level is represented by the metrics significantly contributed to the first four PC components, including *Area*, *Perimeter, Direction*, *Asymmetry*, and *Roundness*.

### *B. The Correlation Between Positional Accuracy Metrics at Per-Building Level*

Two metrics are computed for positional accuracy: distances of corner points, dis(*crn*), and distances of centroids, dis(*ctr*). To analyze the correlation between them, the extracted building result from M4 ("LiDAR+corr") is evaluated as an example. Forty-seven building corner points were ground surveyed by a GPS. These points are matched with the corresponding corners of the extracted buildings on image to calculate corner point positional errors. For the buildings which the corner points were collected, their building centroids are computed and the corresponding distance between the extracted and reference building centroids are calculated. The relationship of the two distance metrics are shown in the scatter plot (Fig. 8) with the two distance axes. The correlation coefficient for these two distance metrics is 0.133. They are not significantly correlated at 0.05 level. It indicates that the corner point's positional change is not significantly related to the centroid positional change. Therefore, the distance metrics on both corner points and building centroids are used to evaluate the *positional accuracy*.

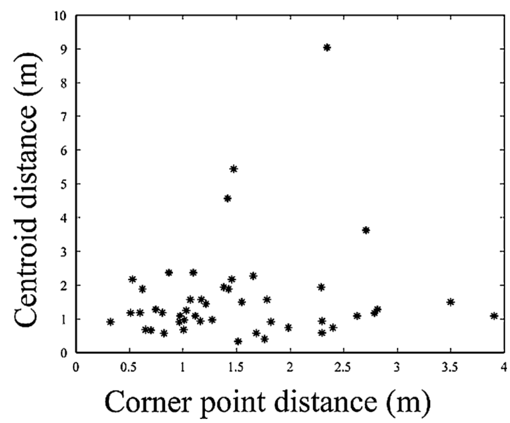


Fig. 8. The scatter plot of the corner point distances and the corresponding building centroid distances for buildings.

### *C. The Aggregation of Duplicate Metrics Based on the Correlation Analysis*

At the *per-scene* level, different metrics are derived from the metrics for individual buildings (such as the mean ratio for *shape similarity*) or computed directly from per-scene level (e.g., the *kappa coefficient* and the *image moment* similarity distance). These metrics are still too many for effective evaluation and there are possible correlations between them. To further compare and integrate these metrics, they are grouped by nine different regions shown in Fig. 4. Using the extracted result from M4 ("LiDAR+uncorr"), all metrics in each region is separately computed and the correlations between the metrics are reported in Table III.

In Table III, $Q$ and *Kappa* represent the *matched rate* and they have very high correlation (0.99). Considering that $Q$ is concisely defined and easy to compute, *Kappa* is discarded in the upper level of evaluation. Many ratios for *shape similarity* are correlated to each other, especially the ratio for *perimeter*, *asymmetry*, *main direction*, and *roundness*. The r(*Area*) and *Img_mnt*

TABLE III
THE CORRELATIONS BETWEEN THE METRICS AT *PER-SCENE* LEVEL

| | *Q* | *Kappa* | *r(Area)* | *r(Peri)* | *r(Asym)* | *r(Dir)* | *r(Rnd)* | *Img_mnt* | *M(ctr)* |
|---|---|---|---|---|---|---|---|---|---|
| *Q* | 1.00 | | | | | | | | |
| *Kappa* | **0.99** | 1.00 | | | | | | | |
| *r(Area)* | 0.48 | 0.44 | 1.00 | | | | | | |
| *r(Peri)* | **-0.68** | **-0.67** | -0.47 | 1.00 | | | | | |
| *r(Asym)* | **0.88** | **0.87** | 0.56 | -0.60 | 1.00 | | | | |
| *r(Dir)* | **0.78** | **0.76** | 0.43 | **-0.96** | **0.71** | 1.00 | | | |
| *r(Rnd)* | **0.67** | 0.62 | 0.56 | -0.44 | **0.77** | 0.56 | 1.00 | | |
| *Img_mnt* | 0.20 | 0.27 | -0.59 | -0.04 | 0.00 | 0.13 | -0.06 | 1.00 | |
| *M(ctr)* | 0.43 | 0.40 | 0.27 | **-0.68** | 0.29 | **0.70** | 0.45 | 0.07 | 1.00 |

Note: the correlation coefficients in **bold** indicate that they are significant at 0.05 level.

are not significantly correlated to any others at 0.05 level. Therefore, the four correlated ratios are combined as only one single metric with equal weight, called "r(*Other*)". At the upper level, three metrics are used to represent the *shape similarity*: r(*Area*), the combined r(*other*), and the *Img_mnt*. Although correlation matrices can vary for different study cases, the strategy to assess correlations and congregate them into relatively independent metrics is the same.

After the selection of metrics at *per-scene* level, there are less metrics left. For *matched rate*, $Q$ is selected; for shape similarity, r(*Area*), r(*Others*), and *Img_mnt* are selected; for positional accuracy both the distances of building centroids and corner points are selected.

### D. Summary of Metrics to the Overall Level

In the case when many building extraction methods need to be compared, the selected metrics at *per-scene* level can be further summarized as a single index. This summary, however, is subjective because it allocates weights for the three groups of the metrics. But a single evaluation index is the best way to compare various evaluation methods. This summary, though subjective, is valuable and feasible if the application purpose is clear and the experiments are conducted before giving weights to each metric group.

Two issues need to be considered in the summary. First, weights need to be assigned. For an application concentrated on classification accuracy, higher weight should be assigned to the *matched rate* because it is the most important criteria to evaluate the result; for a 3D re-construction application, the weight for *shape similarity* should increase, because *shape similarity* is important to determine the appearance of buildings. Currently positional accuracy is still not a major concern by many applications; usually lower weight can be given to this component. However, with more accurate extraction results and more strict demand in future applications, the positional accuracy will also become important. In this study, the *matched rate* (represented by $Q$) is given a weight of 0.5; the *shape similarity* (represented by r(*Area*), r(*Others*), and *Img_mnt* with the equal weight of 1/3) is given a weight of 0.3; the positional accuracy (represented by dist(*crn*) and dist(*ctr*) with the equal weight of 1/2) is given a weight of 0.2.

TABLE IV
THE METRICS AT *PER-SCENE* LEVEL AFTER CONGREGATION

| Method | *Quality* | r(*Area*) | r(*Other*) | *Img_ mnt* | dist (*ctr*) | dist (*crn*) | *Stndzd* Img_mnt* | *Stndzd* dist(*ctr*) | *Stndzd dist(crn)* |
|---|---|---|---|---|---|---|---|---|---|
| M1 | **0.41**+ | 0.69 | 0.65 | 40.25 | 4.95 | 4.47 | 0.41 | 0.41 | 0.41 |
| M2 | 0.70 | 0.77 | 0.80 | 3.16 | 2.20 | 2.38 | 0.84 | 0.75 | 0.73 |
| M3 | 0.79 | 0.85 | 0.76 | 4.90 | 1.54 | 2.51 | 0.82 | 0.84 | 0.71 |
| M4 | 0.80 | **0.86**# | 0.83 | 1.44 | 1.36 | 1.53 | 0.86 | 0.86 | 0.86 |

Note: M1: CIR image; M2: LiDAR; M3: LiDAR+uncorrCIR; M4: LiDAR+ corrCIR.
* *Stndzd* : "standardized"; "+": the minimum metric value; "#":the maximum metric value.

The second issue is the strategy to normalize various metrics. The most commonly used strategy is to rescale the distance metrics to a range of 0 and 1, as follows:

$$\mathrm{x}' = (\mathrm{x} - \min(\mathrm{X}))/(\max(\mathrm{X}) - \min(\mathrm{X})) \tag{2}$$

where X is the vector of all samples and x is an individual sample. *min* and *max* are the minimum and maximum values in the samples, respectively. $\mathrm{x}'$ is the new normalized metric between 0 to 1. However, this method sets the worst extraction method with a new metric value of 0, while the best extracted method will receive a new metric value 1. This may not be fair for metrics with different value ranges. As a result, an improved normalization changes the range as follows:

$$\mathrm{x}' = \mathrm{B} + (\mathrm{A} - \mathrm{B}) \times (\mathrm{x} - \min(\mathrm{X}))/(\max(\mathrm{X}) - \min(\mathrm{X})) \tag{3}$$

where A and B are introduced as the new range. A and B are the *minimum* and *maximum* values of all *ratio* metrics at current level, respectively. That is to say, the range of 0 to 1 is replaced by the current range of ratio metric values. For the distance metric, lower values correspond to better extraction performance. Equation (3) has reversed the distance metric so that the higher value represents higher accuracy and better performance. This is consistent with other accuracy measures. Finally the overall summarized index can be calculated by (1) to compare the performances of all methods.

## VII. RESULTS AND DISCUSSIONS

### A. Performance Assessment by the Evaluation System

Based on the buildings extracted by the four methods introduced in Section V, different metrics are computed for comparison. The metrics for each level are computed according to the evaluation system described in Fig. 3. At *per-building* level, PCA is used to select representative metrics from *shape similarity* metrics. At *per-scene* level, metrics from the three different components are further congregated based on the correlation analysis. The non-ratio metrics (e.g., centriod distance) are normalized to make them comparable with the ratio metrics. The metrics at *per-scene* level is reported in Table IV. With example weights for a general application (*match rate*: 0.5, *shape similarity*: 0.3, *positional accuracy*: 0.2), an *overall* summarized index for the building extraction results is reported in Table V.

From the perspective of *matched rate* described as *Quality* in Table IV, the accuracy from M1 to M3 is gradually improved. This improvement indicates that the method with only CIR

TABLE V
THE SUMMARIZED INDEX AT *OVERALL* LEVEL BASED ON THE THREE COMPONENTS

| Method | Matched rate | Shape similarity | Positional accuracy | **Summarized index** |
|---|---|---|---|---|
| M1 | 0.41 | 0.58 | 0.41 | **0.46** |
| M2 | 0.70 | 0.80 | 0.74 | **0.74** |
| M3 | 0.79 | 0.81 | 0.77 | **0.79** |
| M4 | 0.80 | 0.85 | 0.86 | **0.82** |

image has the worst *matched rate*, while LiDAR can significantly improve the extracted accuracy. The combination of LIDAR with uncorrected CIR images slightly improves the extraction accuracy. M3 and M4 have very similar accuracy on *matched rate*, while they out-performed M1 and M2. From the perspective of *shape similarity* and *positional accuracy*, the four methods show similar trend. From M1 to M4 in Table IV, the ratio metrics increase from the first to last, while non-ratio metrics show largest distance for M1 and decrease from M1 to M4.

It is difficult to compare extraction methods in Table IV because of too many metrics; therefore, a single metric for *shape similarity* in Table V provides straightforward information for users to differentiate extraction methods. In contrast to *matched rate*, one noticeable difference is *shape similarity* and *positional accuracy* metrics in Table V show clearly that M4 performs much better than M3.

The *summarized index* indicates that the performances of the four methods are improved from M1 to M4 in Table V, which is consistent with detailed metrics in Table IV and image visual assessment in Fig. 6. The method with only aerial imagery (M1) performed much worse than the other three methods; with LiDAR data (M2), the extraction can improve remarkably; while the combination of these two data (M3) slightly improved the performance. The difference for methods with LiDAR and aerial imagery before (M3) and after (M4) relief correction can be identified in the *summarized index*, while such a subtle difference cannot be pointed out by merely *matched rate*. This conclusion is consistent with the visual evaluation of Fig. 6.

### B. The Comparison of Evaluation Systems: Traditional Method Versus the Proposed System

The comparison of the four building extraction methods on the entire study site has demonstrated that the evaluation system can provide consistent assessment with human visual evaluation. But individual buildings in the overview maps of Fig. 6 are small and difficult to perceive. As a result, a few example buildings are compared between M3 and M4 in this section.

To analyze the performance of the proposed evaluation system, traditional methods based on the classification accuracy are used to compare with the proposed evaluation system. As introduced in Section III, such traditional methods use *matched rate* metrics such as *Completeness, Correctness, Quality*, etc. Considering *Quality* is a popular metrics for evaluation and it has been applied in recent studies [1], [7], it is used as the traditional evaluation method to compare with the proposed evaluation system in this study.

Four sample buildings extracted from M3 and M4 are analyzed. In Fig. 9, extraction results from the two methods are overlaid on the reference buildings. Extracted buildings from the two methods are similar to each other, except that the results from M4 have neater boundaries. The improvement from M3 to M4 is mainly on the image segmentation stage, where M3 will generate more fragmented buildings because vertical LiDAR data sometimes conflict with the tilted buildings on the aerial image before relief displacement correction.

In Fig. 9, human visual evaluation can easily identify that building boundaries from M3 are not well matched with the reference buildings. The traditional method with *Quality*, however, is not sensitive to the fluctuation on the border; the *Quality* difference between M3 and M4 is less than 0.05 in Table VI. In contrast, *shape similarity* metrics can detect such difference, with reported 15% to 30% improvement from M3 to M4 (Table VI). Case (c) in Fig. 9 is an extreme case, where *Quality* is coincidently the same between M3 and M4. The extracted building from M3 is obviously less accurate than that of M4 compared with the reference building footprint in (c). It is the other components (*shape similarity* and *positional accuracy*) that can distinguish the two results. The *overall* summarized index for these buildings can differentiate the extraction accuracy that cannot be separated by the traditional method. Compared with the negligible $Q$ difference between M3 and M4, our evaluation system shows considerable differences in *shape similarity* and *positional accuracy*.

It can be noticed that the performance of metrics at *per-building* level in Table VI is better than at per-scene level in Table IV. That is because *per-scene* level needs to consider “zero-to-one” and “one-to-zero” cases. The buildings fail to be matched will decrease the overall accuracy.

### C. Discussion of Building Types Via the Multi-Level System

The multi-criteria and hierarchical evaluation system with three levels provides a framework of metrics. Such a multi-level system is valuable for tracing metrics from top to bottom. For example, a low *overall* index can be searched downward in order to identify the reason. It may be owing to the low shape similarity at *per-scene* level; the low *shape similarity* can be further investigated, it may be caused by low perimeter ratio at *per-building* level. Such an analysis provides feedback to further improve building extraction methods. Furthermore, metrics at the bottom level (per-building) can be sorted, grouped, or clustered, in order to discover the distribution of extraction errors. For instance, sorting extracted buildings according to a metric at *per-building* level can assist manual editing, to firstly modify the buildings with the lowest metric value. Moreover, grouping extracted buildings with a certain attribute can test systematic bias related to different types of buildings.

As an example, building’s occupancy is investigated to analyze whether an extraction method is effective at a given building type. In Table VII, buildings are grouped as two types: public buildings and residential buildings; therefore, metrics at

TABLE VI
THE CORRESPONDING METRICS OF THE FOUR EXAMPLE BUILDINGS

| | Area ($m^2$) | **Traditional method (Quality)** | | | Shape similarity* | | | Center distance (m) | | **This study (overall)** | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ID | | M3 | M4 | **diff^** | M3 | M4 | **diff** | M3 | M4 | M3 | M4 | **diff** |
| (a) | 4973 | 0.88 | 0.93 | **0.05** | 0.66 | 0.96 | **0.30** | 1.27 | 0.69 | 0.78 | 0.94 | **0.15** |
| (b) | 6069 | 0.92 | 0.96 | **0.04** | 0.73 | 0.98# | **0.25** | 1.54 | 1.25 | 0.80 | 0.92 | **0.12** |
| (c) | 227 | 0.79 | 0.79 | **0.00** | 0.62+ | 0.85 | **0.23** | 1.46 | 1.01 | 0.71 | 0.81 | **0.10** |
| (d) | 881 | 0.83 | 0.83 | **-0.01** | 0.80 | 0.94 | **0.14** | 0.93 | 0.53 | 0.82 | 0.89 | **0.07** |

Note: +the minimum metric value; #the maximum metric value; ^ diff=M4-M3; *shape similarity at per-building level has no image moment distance, which has been replaced by perimeter ratios.

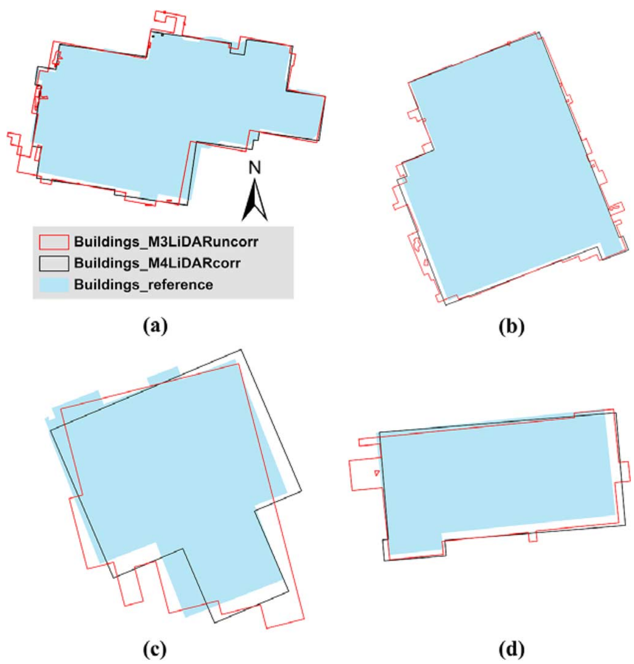


Fig. 9. Four examples of the extracted buildings by two extraction methods (M3 and M4).

TABLE VII
THE METRIC COMPARISON BETWEEN PUBLIC BUILDINGS AND RESIDENTIAL BUILDINGS

| | Size($m^2$) | *Comp* | *Corr* | *Q* | *kappa* | dist(*ctr*) |
|---|---|---|---|---|---|---|
| Pub* | 1420 | 0.94 | 0.89 | 0.84 | 0.90 | 1.48 |
| Res# | 150 | 0.74 | 0.87 | 0.67 | 0.79 | 1.32 |
| | r(*Area*) | r(*Peri*) | r(*Asym*) | r(*Dir*) | r(*Rnd*) | *Img_mnt* |
| Pub | 0.89 | 0.77 | 0.87 | 0.91 | 0.83 | 0.17e-3 |
| Res | 0.85 | 0.85 | 0.78 | 0.79 | 0.79 | 0.39e-3 |

Note: * Regions 1-3 in the study area is grouped as public buildings (Pub); #regions 4-9 are grouped as residential buildings (Res).

*per-building* level are grouped to re-calculate their *per-scene* metrics based on their types.

Metric values of public buildings are generally higher than that on the residential buildings. The public buildings are usually larger and higher (with less tree covering) than residential buildings; thus they are easier to be detected. Two metrics are very different for large buildings and smaller ones: the distance between centroids and the ratio for perimeter. The centroid locations of larger buildings with complicated shapes can be affected when one corner is missed. The residential buildings are usually rectangles. Therefore the centroid locations do not shift much between the extracted and the reference buildings. This analysis is important to improve the advanced algorithm design and to better detect buildings in residential areas.

## VIII. CONCLUSION

This study proposes a multi-criteria and hierarchical evaluation system for building footprint extraction methods. After a review of the current methods and strategies used in building extraction evaluation, the problems of current evaluation methods are summarized as: overestimating the importance of *matched rate*, incomplete description of *shape similarity*, overlooking the redundancy between metrics, and lack of an *overall* index for comparison purpose. Consequently, a multi-criteria evaluation system represented by three components from different perspectives is proposed. These components are: (a) *matched rate* that describes the proportion of the building area successfully extracted when compared with the reference data, (b) the *shape similarity* that describes to what extent an extracted building is similar to the reference, and (c) the *positional accuracy* which evaluates the geometric distance between extracted and reference buildings.

The proposed evaluation system is also stratified into three levels: the *per-building* level, *per-scene* level, and the *overall* level. The *per-building* level evaluates individual buildings with different metrics; it can provide detailed information for each building and it is valuable for manual editing after extraction. The *per-scene* level treats each scene as a unit for evaluation and different metrics are provided to assess the whole scene. It is the major level for most evaluations. The *overall* level tries to summarize different metrics as a single index in order to perform an integrated comparison of different building extraction methods. Although the *overall* level is summarized by setting weights subjectively, it provides a concise manner for comparison. Future work may discuss how to obtain reasonable weights based on reliable survey or experiments.

The proposed evaluation system is tested by four building extraction methods. The four methods use different data sources, including the original CIR images, LiDAR data and their combination. The last method combines LiDAR data with CIR image after relief displacement correction. These four methods are implemented and compared. The system demonstrates its superiority to traditional evaluation methods, especially when traditional classification accuracy cannot distinguish the performance of two extraction methods, while the proposed system can identify the best performer very well. Finally, the impact of building types on evaluation methods is also analyzed by taking advantage of this multi-level system.

In summary, this study explores the evaluation methods to assess building extraction accuracy from remotely sensed imagery. Specifically, (a) to complete the description about the *shape similarity* metrics, twelve shape metrics are developed at the *per-building* level and principal component analysis is conducted to remove redundancy between metrics; (b) To remove correlations between different metrics, correlation analysis at the *per-scene* level are utilized to select and aggregate metrics. Aggregated metrics simplifies the comparison at the *per-scene* level with less relatively independent metrics. (c) To make the comparison of methods straightforward, an overall index is proposed by using the weighted linear combination of the three evaluation components. This study also provides a way to integrate ratio and distance metrics by rescale the latter. (d) To test the proposed evaluation system, four different building extraction methods are implemented and evaluated by the proposed system. Evaluation method from this study performs more consistently with visual inspection than traditional evaluation methods.

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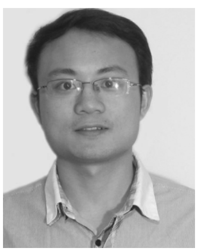

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